



Parcel Post and Express Post

Label & Barcode Specification

Version 1.4
05/06/2025



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Revision History

Version	Date	Author	Changes Made
1.4	05.06.2025	eCommerce Integration Support	• Changed document title • Added "Revision History" • Updated support contact details to remove customer connectivity email address and replace with Lodgement Tech Support form URL/hyperlink • Added Service Code 49 for Wine and removed all Express Wine

Table of Contents

The Purpose of This Document	5
Intended Audience	5
Assumed Knowledge	5
Help Contact Details	5
eParcel Integration Requirements	5
Merchant Location ID (MLID)	6
Delivery Point Identifier (DPID)	6
Manifest, Consignments and Articles	6
Relationship between Orders (Manifests), Shipments (Consignments) and Items (Articles)	7
Example of an Order (Manifest) with Shipments (Consignments) and Items (Articles)	7
The eParcel label	8
Traditional Parcel Post label layout	8
Contemporary Parcel Post label layout	9
Contemporary Express Post label layout	10
Example of a Parcel Post label (15 by 10cm)	11
Example of an Express Post label (15 by 10cm)	12
Example of an eParcel Parcel Post Return label	13
Check List for Parcel Post & Express Post label contents and format	14
Parcel Post/Express Post Barcodes	15
The Parcel Post/Express Post GS1-128 Linear Barcode	15
Example of a domestic eParcel GS1-128 barcode.	15
Component descriptions of domestic eParcel GS1-128 Linear Barcode	17
GS1 Datamatrix Barcode	18
Example of a domestic eParcel GS1 Datamatrix Barcode.	18
Component descriptions of domestic eParcel GS1 Datamatrix	19
Service Codes, Product Codes and Product_ids	20
Service options for eParcel	21
The eParcel Check Digit	22
Reference table for converting alpha characters into digits	22
Example 1	22
Example 2	23
Check Digit calculation - Sample code (VBA)	23
Symbol Check Character	23
Options for label printing	24
Label and barcode testing	25
Testing for quality with the barcode Verifier	25
SSCC barcodes	26
International labels	26
Preliminary “in house” barcode testing	27
Regular label quality checks and damaged labels	29
Label placement issues	30
Tracking details via API	31

The Purpose of This Document

This document explains contents and formatting of Australia Post Parcel Post / Express Post labels and barcodes. Australia Post, to provide an enhanced experience and visibility to its customers, has invested in automation of parcel processing and has equipped all its delivery agents with handheld scanning devices. Overall success of course is heavily dependent on parcel presentation (labelling and barcoding) as well as accuracy and timeliness of electronic Order (manifest) data.

Intended Audience

This document is intended for:

- Merchants currently using or intending to use Parcel Post or Express Post products generating barcoded labels from their own system.
- Merchants' Information Technology staff who are involved in developing Parcel Post or Express Post barcoded labels.
- Australia Post representatives supporting their merchants during the systems integration process via our Shipping & Tracking APIs.

Assumed Knowledge

The following concepts are assumed knowledge:

- GS1 numbering and symbology (GS1 2D Datamatrix barcode and GS1 128 linear barcode)
- Technology that helps and supports building those barcodes.

Help Contact Details

If you have questions or concerns regarding charges to your account, please contact your Account Manager. For queries relating to integration via our Shipping & Tracking APIs or our shipping labels, please refer to our [Developer Centre](#) which contains resources and technical documentation on our APIs.

If further assistance is required, please raise a case through our [Lodgement Tech Support form](#). Please ensure that you provide the account number, API key, payload and error you have experienced as this detail is required to speed-up the investigation.

eParcel Integration Requirements

The following is a summary of what is required to integrate with our Lodgement systems:

- Produce Parcel Post / Express Post barcoded labels and get these tested and approved by Australia Post. (See sections 3 and 4 for further details)
- All labels and electronic data (Orders/manifests) produced from merchants' own systems must be fully tested and approved before use. Please access this testing service through your eCommerce Integration Specialist who will provide you with the "TechSignOff" customer form for you to complete.

In addition, to complete the integration piece, merchants need to provide the following (not part of this document):

- Produce a hard copy manifest (Order Summary) and get it approved by Australia Post. This is not required for returns.

We strongly recommend Merchants source the Order Summary from the Shipping & Tracking APIs.

- Produce and transfer an electronic manifest (visit the Australia Post Developers Centre for information <https://developers.auspost.com.au/apis/shipping-and-tracking/reference>)

For label testing, **please ensure that you use your existing assigned production MLID**. If you do not yet have a production account or MLID, then the following details can be used.

- o Merchant Location Id JDQ (3 characters MLID)
- o Merchant Location Id 1JDQ1 (5 characters MLID)

Once you pass all tests you will be provided with production details which include:

- Charge account number
- Username (API key)
- API Secret

Please contact your eCommerce Integration Specialist if assistance is required.

Merchant Location ID (MLID)

A Merchant Location ID (MLID) is a three or five alphanumeric value which represents an account/sub-account number, and it is linked to a specific account number by default. Depending on your requirements, Australia Post may provide you with multiple MLIDs/account numbers.

A MLID followed by seven digits becomes a consignment number, management of consignment numbers is responsibility of the merchant. Australia Post requires that each individual consignment number is used only once during a minimum period of two years.

It is recommended that you start the number range at 2000001, to allow 2,000,000 consignment numbers to be generated from eParcel Online before encountering the risk of duplicating consignment numbers. Duplication of consignment numbers within two years is a serious issue and will lead to data integrity issues, reporting inaccuracy, and/or charging errors for which Australia Post cannot be held responsible.

Delivery Point Identifier (DPID)

It is an eight digit number that represents a physical address, it is very important to have DPID as part of the barcode and electronic manifest data as it allows deep sort during automatic parcel processing.

DPID is obtained during address validation but not having it in the barcode or manifest does not prevent merchants for lodging parcels. If the DPID section of the barcode string is not included in the barcode, this is still valid.

You can find information about AMAS (Address Matching Approval Systems) at...

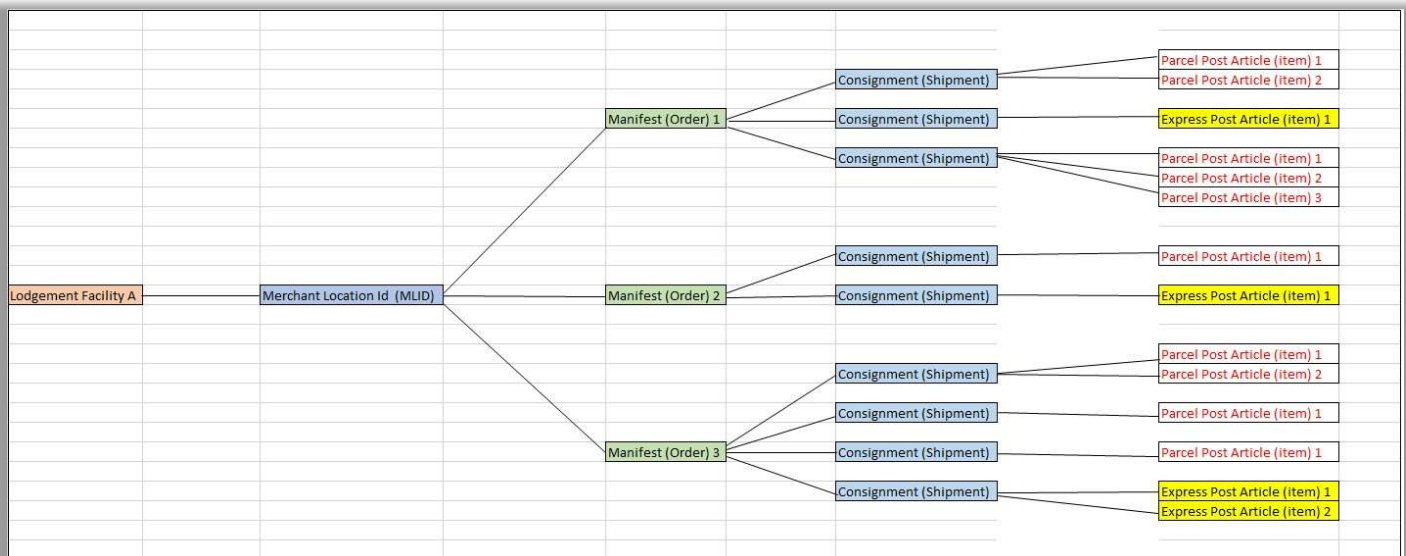
<https://auspost.com.au/business/shipping/letters-domestic/large-volume/mail-barcoding/address-matching-approval-system-amas>

Manifest, Consignments and Articles

- An Order (a manifest) is a container for all consignments in each lodgement.
- A Shipment (a consignment) can have one or more items (articles) in a parent-child relationship, the maximum number of items (articles) in a shipment (consignment) is 20.

The Australia Post smart barcode embeds the consignment number within the article number, this means that all articles in a consignment share the MLID and the subsequent 7 digits and all articles need to be included in the same order (manifest).

Relationship between Orders (Manifests), Shipments (Consignments) and Items (Articles)



Example of an Order (Manifest) with Shipments (Consignments) and Items (Articles)

Order (manifest) **AP000000001** contains two shipments (consignments), 1JDQ12000001 and 1JDQ120000002

Shipment (Consignment) **1JDQ120000001** is Parcel Post (plus Signature) and contains 20 articles (the maximum for a consignment) with delivery address 1 George St Sydney NSW 2000, the sender requested signature on delivery unless the receiver requests it to be safe dropped.

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The eParcel label

There are two main label layouts that can be used as reference for those customers who choose to generate labels in house, namely:

- Traditional with a horizontal linear barcode from left to right
- Contemporary with a vertical linear barcode from top to bottom.

There are pros and cons

- The contemporary format with its vertical linear barcode prevents printer errors (faulty heating element for example) from cancelling the whole barcode however if the label is of a PDF format printed on thermal printers that is a bad combination which results in an unreadable barcode.
- In the other hand the Traditional label template presents a clear horizontal barcode to scanning devices, but it is vulnerable to printing errors due to dirty printer drums, faulty heating elements or defective label stationery which are noticeable as blank lines from top to bottom of the label, these distort the barcodes (causing “no reads” as a result).

You can implement either of the two label layouts but consider the one that best suits the technique being used. There are features that are non-negotiables like the need for both GS1 compliant barcodes, the linear GS1 128 barcode and the 2D GS1 Data Matrix symbol.

The “Parcel Post” Label can be printed in black and white however the “Express Post” label requires the distinctive yellow Express Post colour in the label header. This is so required for “product” identification at lodgement so the product can be streamed into the speedy network for overnight delivery.

The sender can take one of the following options to comply with the requirement for colour on Express Post labels:

- The sender can source similar stationery from their preferred supplier (at the sender’s own cost). This may imply a dedicated printer for Express Post labels.
- The sender could print Express Post labels in black and white and attach an additional coloured Express Post tape to the parcel (this Express Post tape is available free of charge at the nominated Australia Post Lodgement Facility)

Traditional Parcel Post label layout

The example shows a 3 characters MLID (JDQ)

Parcel Post

DELIVER TO PHONE 0415446586

Test
Company Name
Address Line 1
Address Line 2
Address Line 3
Address Line 4
CHULLORA NSW 2190

DELIVERY INSTRUCTIONS .84kg

SIGNATURE ON DELIVERY REQUIRED CON NO JDQ0194571
PARCEL 1 OF 1

AP Article Id: JDQ019457101000930308

AP Article Id: JDQ019457101000930308 0525

SENDER
Test
This is A Test Only
MELBOURNE VIC 3000

Aviation Security and Dangerous Goods Declaration
The sender acknowledges that this article may be carried by air and will be subject to aviation security and clearing procedures, and the sender declares that the article does not contain any dangerous or prohibited goods, explosive or incendiary devices. A false declaration is a criminal offence.

Date of label generation (MMDD) goes here

Contemporary Parcel Post label layout

This example shows a 5-character MLID (12345)



Parcel Post

To:

Customer Name
 Company Name
 Address Line 1
 Address Line 2
 Address Line 3
 CHULLORA NSW 2190

Postage Paid



This is a test label only not to be used in production

Ph : 0397654321

Dead weight Delivery features

2.00kg

Cons No: 123451234567
 Parcel 1 of 1

From:

Sender Pty Ltd
 Sender's Address Line 1
 MELBOURNE VIC 3000

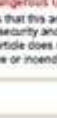
Aviation Security and Dangerous Goods Declaration

The sender acknowledges that this article may be carried by air and will be subject to aviation security and clearing procedures, and the sender declares that the article does not contain any dangerous or prohibited goods, explosive or incendiary devices. A false declaration is a criminal offence.

Ref-1: VALUE-1

Ref-2: VALUE-2

AD 4434316 123451234567000109003 0911



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Note: The arrows indicate the position where the date (MMDD) of label generation needs to be printed

Contemporary Express Post label layout

The Express Post label layout is similar to the Parcel Post label, just the Express Post label header (product branding) needs to be used instead.

Colour is critical on this label so if label prints in black and white it is a valid approach to attach the additional Express Post tape (available at your nominated lodgement facility).

The size of the Express Post label is 15 cm by 10 cm regardless the layout (traditional or contemporary).

Express Post

To:
Greg E Burns
123 Nott St
PORT MELBOURNE VIC 3007

Postage Paid

This is a test label only not to be used in production

Ph : 0397654321

Dead weight **2.00kg** Delivery features

From: Cons No: 123451234567
Parcel 1 of 1

SupaCompressors
Shay Healey
124 Northy St
CHATSWOOD NSW 3000

Aviation Security and Dangerous Goods Declaration
The sender acknowledges that this article may be carried by air and will be subject to aviation security and clearing procedures; and the sender declares that the article does not contain any dangerous or prohibited goods, explosive or incendiary devices. A false declaration is a criminal offence.

Ref:1: VALUE-1

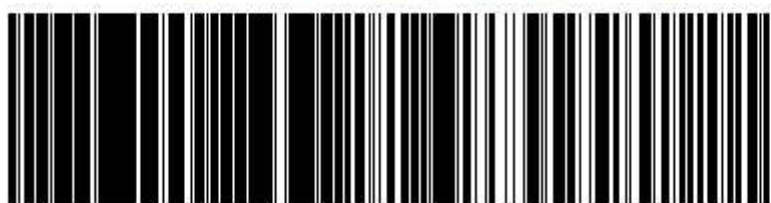
Ref:2: VALUE-2

AP Article ID: 123451234567010008709500

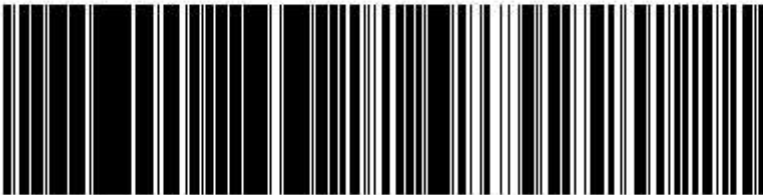
0611

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Note: The red arrow in the label sample indicates the position where the date (MMDD) of label generation needs to be printed

 Parcel Post	
DELIVER TO Stephen Atkinson 47 Smith St LORNE VIC 3232	PHONE 0352895288 
DELIVERY INSTRUCTIONS .44kg	
IF PREMISES INATTENDED, LEAVE IN A SECURE LOCATION OUT OF THE WEATHER	
AUTHORITY TO LEAVE IF UNATTENDED	CON NO 2AP0670948 PARCEL 1 OF 1
AP Article Id: 2AP067094801000930800	
	
AP Article Id: 2AP067094801000930800 0625	
SENDER Apl Po Box 6003 CHULLORA NSW 1405	Aviation Security and Dangerous Goods Declaration The sender acknowledges that this article may be carried by air and will be subject to aviation security and clearing procedures; and the sender declares that the article does not contain any dangerous or prohibited goods, explosive or incendiary devices. A false declaration is a criminal offence. 8824109968 0394183777

← Date of label generation (MMDD)

		Express Post	
DELIVER TO Stephen Atkinson 47 Smith St LORNE VIC 3232		PHONE 0352895288 	
DELIVERY INSTRUCTIONS		.44kg	
IF PREMISES INATTENDED, LEAVE IN A SECURE LOCATION OUT OF THE WEATHER			
AUTHORITY TO LEAVE IF UNATTENDED		CON NO 2AP0670948 PARCEL 1 OF 1	
AP Article Id: 2AP067094801000960807			
			
AP Article Id: 2AP067094801000960807		0625 	
SENDER Apl Po Box 6003 CHULLORA NSW 1405		Aviation Security and Dangerous Goods Declaration The sender acknowledges that this article may be carried by air and will be subject to aviation security and clearing procedures; and the sender declares that the article does not contain any dangerous or prohibited goods, explosive or incendiary devices. A false declaration is a criminal offence. 8824109968	

Example of an eParcel Parcel Post Return label

Parcel Post	
RETURNS	
Post Office Use only: Scan barcode, no payment required. Where no printed label presented, scan either barcode and select "Y" to print labels or access T/C 3891 and enter	
DELIVER TO The Shop Po Box 144 CHULLORA NSW 1405	
DELIVERY INSTRUCTIONS	
SIGNATURE ON DELIVERY REQUIRED PARCEL 1	
AP Article Id: 33334007731201000650309	
	
AP Article Id: 33334007731201000650309	
SENDER Test Customer 555 Warringah Road FORESTVILLE NSW 2087	Aviation Security & Dangerous Goods *I the sender acknowledge that this article may be carried by air and will be subject to aviation security and clearing procedures, and I declare that it does not contain any dangerous or prohibited goods, explosives or incendiary devices. If the declaration is not signed this article cannot be carried by air. Please sign here:
Reference 1:REFERENCE 1	
Reference 2:REFERENCE 2	

INSTRUCTIONS
1. Check
Please ensure the print quality of the label is clear and legible. If you do not have access to a printer please refer to point **4.Print at Post**. If you are sending multiple parcels, a separate label with a unique barcode is required for each parcel.
2. Sign & Secure
Sign the Aviation Security and Dangerous Goods Declaration and securely attach the label to the top face of the parcel, ensuring tape does not cover any part of the barcode. Please pack your item securely.
3. Post
Post your parcel at any Post Office and you will be issued with a lodgement receipt. Alternatively, post your parcel in a red street posting box or in-store drop box (if size permits and your retailer has not specified otherwise).
4. Print at Post
You may print your label at most Post Offices - see auspost.com.au/print-locations to find out which Post Offices are included. You should:

- **Either** bring an electronic copy of this label on your phone or tablet and present this to the Australia Post counter staff who will scan the barcode print the return label and attach it to your parcel
- **Or** quote the following number to the Australia Post counter staff to enable them to print and attach a return label

Return Reference no. **333340077312**

Note: the resolution of the eParcel barcodes in figures 3, 4 and 5 are not suitable for quality test, the aim of this figure is to show the standard label layout only.

Check List for Parcel Post & Express Post label contents and format

The following table should be used as reference to evaluate whether the Parcel Post/Express Post label format is likely to cause issues with the Optical Recognition Software (ORS).

Requirement	Mandatory/Desirable	Comment
White label with black font	Mandatory	No colour background in the address sections or barcode section
Label is on 1 flat surface	Mandatory	Label is not wrapped across multiple faces of the parcel
Standard label template	Highly Desirable	The standard label templates are configured in the ORS, so it looks for the address in these positions
To Address		
The address block is 1 single block of text	Mandatory	To: MR C Receiver Unit 14 1 Test Street CHULLORA NSW 2190
Address block should be left justified	Mandatory	
Address should be in the top left section	Mandatory	
Font size is consistent within the address block	Mandatory	Regular 12 to 14 Arial font (or similar)
Font size is Arial 12	Highly Desirable	Font size below 10 will provide a poorer result
No truncating of address information	Mandatory	e.g. suburb should not be truncated The ORS uses the full address not just the post code
Keyword DELIVER TO or TO use at the top of the address block	Mandatory	No line spacing or lines between keyword and address block
Text including the keyword is in normal font i.e. not bold or italics	Mandatory	
No lines or gaps (blank lines or excessive spacing) within the address block	Mandatory	
Suburb, State and Post code to be on 1 line and capitalised with no commas	Mandatory	e.g. MELBOURNE VIC 3001
Address is a valid address that complies with Australian addressing standards	Highly Desirable	The use of address validation software would improve the accuracy of addresses
From Address		
The address block is 1 single block of text	Mandatory	from: MR C SENDER Unit 1 1 Test Street CHULLORA NSW 2190
Address block should be left justified	Mandatory	
Address should be in the bottom left section	Mandatory	
Font size is consistent within the address block	Mandatory	
Font size is Arial 10	Highly Desirable	Return address should be smaller than TO address. In the future the return address will be used by video coding, so needs to meet ORS standards
No truncating of address information	Mandatory	It is desirable to shrink the text rather than truncate information
Keyword FROM or SENDER use at the top of the address block	Mandatory	
Text including the keyword is in normal font i.e. not bold or italics	Mandatory	
No lines or gaps (blank lines or excessive spacing) within the address block	Mandatory	
Suburb, State and Post code to be on 1 line and capitalised with no commas	Mandatory	
Dangerous Goods Declaration	Mandatory	If the article contains small quantities of DGs (if allowed in the merchant contract) then the DG declaration changes to "FOR ROAD TRANSPORT ONLY" and DG stickers need to be applied to the carton.
Address is a valid return address	Mandatory	
Date time (MMDD) of label generation in the expected position	Mandatory	

Parcel Post/Express Post Barcodes

Australia Post barcodes and numbering are in line with GS1 standards, two barcodes are required on domestic parcels to assist Australia Post in providing the best possible service, namely:

- **A linear GS1-128 barcode.**

Used for article tracking on devices unable to scan a 2D Datamatrix barcode and containing:

- Australia Post GTIN
- Article Tracking Reference

- **A two-dimensional GS1 Datamatrix Barcode**

Used for Article Tracking and containing the same information as the GS1 Linear Barcode above but also including additional components:

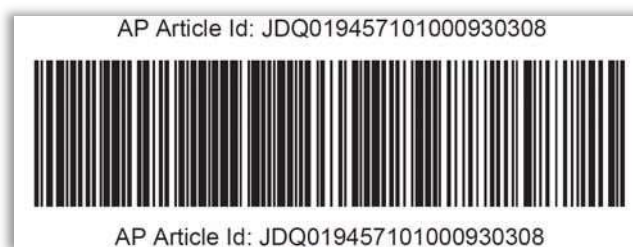
- Australia Post GTIN
- Article Tracking Reference
- Postcode
- Delivery Point Identification Number (DPID)
- Barcode Production Date/Time

The Parcel Post/Express Post GS1-128 Linear Barcode

- Barcode width: 83.8mm (3 characters MLIDs) or 89.4mm (5 characters MLIDs).
- Barcode Height: Recommended 22mm (18 mm minimum).
- Human readable: Arial 12. Characters in the human readable line logically arranged, article number preceded by the text "AP Article ID:"...
- Light Margin: 2.54 mm minimum at each side.

Example of a domestic eParcel GS1-128 barcode.

<F1>019931265099999891JDQ019457101000930308



Each alpha or numeric character in a GS1 128 barcode is represented by a combination of 11 dark and light modules arranged in different number sets known as A, B, and C.

In the Parcel Post\Express Post barcode we generally use character sets A and C:

- Set A encodes one character per module and includes all the standard upper-case alpha and numeric characters, plus control and special characters, we use this character set to encode Merchant Location Id which often contains alpha characters.

Note: It is fine to use set B instead of set A, as long as any alpha characters are kept in upper case.

- Set C is for numerical characters (digits) only and includes the set of 100-digit pairs from 00 through 99, as well as three special characters. Set C is the most efficient as it encodes data at twice the density of set A so therefore represents two digits in one symbol character.

Some commercially available barcode production software will use character set B in place of character set A and this is acceptable if capitals are always used.

The eParcel barcode uses the following auxiliary characters:

- start C
- code A
- code C
- FNC1
- Stop

The start character in the eParcel barcode is always Start C as it always begins with the set C (numeric characters).

Code A and C characters allow a change in character set to be affected within the barcode. All characters following the code character are encoded according to the corresponding character set until another code character appears. The following characters are then encoded according to the latest code character. Code A changes the encoding to set A and code C changes it back to set C.

Component descriptions of domestic eParcel GS1-128 Linear Barcode

Description	No. of characters*	Detail
Start C character		
Function 1 Symbol Character		FNC1 (Codeword ASCII value of 232)
Application Identifier	2	01
Indicator	1	9
Company Prefix	7	9312650
Item Reference	5	99999
Check Digit	1	8
Application Identifier	2	91
Code A character		
Merchant Location ID	3 or 5*	The identifier provided by Australia Post will uniquely identify Lodgement locations and merchants may use multiple location IDs.
Code C character		
Consignment ID	7	ID is provided by the merchant. Allows a possible 9,999,999 cons. per Merchant Location before reuse of numbers is required. Numbers must not be reused within a 2-year period.
Article Count	2	Position of the article in the consignment (e.g. 01 is the first, 02 the second etc. up to 20).
Product Code	5	This will be a 2 digit or a 3-digit number padded with zeros. The 2-digit number range is from 63 - 96. The 3-digit number range will be provided by Australia Post. This number defines an Australia Post product
Service Code	2	Defines delivery actions required e.g. 03 signature on delivery always required
Postage Paid Indicator	1	0
Check Digit	1	Used to verify the validity of the eye-readable Article ID
Symbol check character		Used to validate the barcode quality
Stop character		

GS1 Datamatrix Barcode

- GS1 Datamatrix Quiet Zone is one module wide on all four sides. As with other bar code Quiet Zones, do not print in this area.
- The data is encoded using a series of dark or light dots based upon a pre-determined size. The minimum size of these dots is known as the X-dimension. The recommended size of the Datamatrix Barcode is 13.68 x 13.68mm
- The data is then encoded in a matrix within the Finder pattern. This is a translation into the binary Datamatrix Barcodeogy characters (numeric or alphanumeric).
- For GS1 System compatibility, FNC1 shall be encoded at the beginning of the data string and should be used as a group separator. When a FNC1 is used as a group separator, it shall be represented in the transmitted message by the ASCII character <GS> (ASCII value 29).

Characteristics of GS1 Datamatrix are:

- Figure 6 (below) represents a GS1 Datamatrix Barcode with 26 rows and 26 columns (including the perimeter finder pattern but not including Quiet Zones).
- Datamatrix is a matrix (2D or two-dimensional) bar code which may be printed as a square or rectangular symbol made up of individual dots or squares. For eParcel GS1 Datamatrix, only square symbol should be used.
- GS1 Datamatrix solid "L" shaped finder or alignment pattern is one module wide. It is primarily used to determine the size, orientation and distortion of the symbol.
- The other two sides of the finder pattern are alternating light and dark elements, known as the "Clock Track". This defines the basic structure of the symbol and can also help determine its size and distortion.

Example of a domestic eParcel GS1 Datamatrix Barcode.



Full string:

<F1>019931265099999891JDQ019457101000930308<F1>4202190<F1>8008170417155407

Component descriptions of domestic eParcel GS1 Datamatrix

Description	No. of characters*	Detail
Function 1 Symbol Character		Start Character - FNC1 (Codeword ASCII value of 232)
Application Identifier	2	01
Indicator	1	9
Company Prefix	7	9312650
Item Reference	5	99999
Check Digit	1	8
Application Identifier	2	91
Merchant Location ID	3 or 5*	The identifier provided by Australia Post will uniquely identify lodgement locations and merchants may use multiple location IDs.
Consignment ID	7	ID is provided by the merchant. Allows a possible 9,999,999 cons. per Merchant Location before reuse of numbers is required. Numbers must not be reused within a 2-year period.
Article Count	2	Position of the article in the consignment (e.g. 01 is the first article, 02 the second etc. up to 20).
Product Code	5	This will be a 2 digit or a 3-digit number. The 2-digit number range is from 63 - 96. The 3-digit number range will be provided by Australia Post. This number defines an Australia Post product
Service Code	2	Defines delivery actions required e.g. signature on delivery
Postage Paid Indicator	1	0
Check Digit	1	Used to verify the validity of the eye readable Article ID
Function 1 Symbol Character	1	Group Separator – FNC1 (Codeword ASCII value of 232)
Application Identifier	3	420
Postcode	4	Delivery Postcode
Function 1 Symbol Character	1	Group Separator - FNC1 (Codeword ASCII value of 232) - Remove from the barcode string if DPID is not available.
Application Identifier	2	92 (Remove from the barcode string if DPID is not available)
Delivery Point Identifier (DPID)	8	An 8-digit reference which indicates the physical address of the consignment delivery - see Delivery Point Identifier (DPID) Provisioning Remove from the barcode string if DPID is not available.
Function 1 Symbol Character	1	Group Separator - FNC1 (Codeword ASCII value of 232)
Application Identifier	4	8008
Date and Time of label generation	12	YYMMDDHHMMSS

Service Codes, Product Codes and Product_ids

- **The Service Code** is a two-character numeric ID that defines any special delivery conditions for merchants using the Australia Post eParcel barcode. Ones which are relevant to merchants generating their own manifests are:
 - **Signature on delivery always required** - if a signature cannot be obtained at the delivery address, the parcel must be “carded” to a Post Office (service code 03)
 - **Signature on delivery required (safe drop enabled)** – The sender allows the intended receiver of the parcel to opt for ATL by logging into My Post (service code 50)
 - **Authority to leave if unattended** – if there is no one at the delivery address the sender authorizes for the parcel to be left at the premises, appropriate delivery instructions are required, e.g. “leave in a secure place away from weather” (service code 08)

Note: It is important for “Signature on delivery always required” service codes not to be used in conjunction with contradictory Special Delivery Instructions (e.g. “Leave in a safe place”).

- **Partial delivery allowed** - the standard process is to hold articles within a consignment until all articles have arrived at the terminating delivery centre. This service code provides authority to deliver a part consignment.
 - **Returns Service** - eParcel Returns transaction only support “03” or “08” as these are always single parcel consignments
- SSCC numbering system does not allow the embedding of Service Code and Product Code in the barcode so when a merchant is authorized to use SSCC numbers as article Ids we need to set-up a barcode mask in our back-end system so our scanning devices take that number as a valid Parcel Post\Express post article id. Merchant using SSCC numbers can define special delivery conditions at a consignment level by applying relevant values to the following elements in the payload:

```
"authority_to_leave": false/true,  
"allow_partial_delivery": false/true,  
"safe_drop_enabled": false/true,
```

Our system will translate those delivery conditions into service codes and deploy the requirement to scanning devices. If the order (electronic manifest data) is not available at the time of a SSCC label being scanned, the default service associated to the SSCC barcode mask applies.

- **The Product Code** is a five-character numeric ID that defines the eParcel sub-product type. This should be associated with the appropriate product_id (Charge code).
- **The product_id** (also known as charge code) has a direct relationship with the product code but it is not found in the barcode.

It is required in the order (electronic manifest) to manage billing, but it is critical for merchants generating labels in house to know what service codes are available for use against each charge code within their contract.

Service options for eParcel

Make sure that the electronic data does not conflict with the service code in the barcode, use the “APIs payload” column as reference for this.

Service Code	Safe Drop	Signature on Delivery required	Authority To Leave (ATL)	Partial delivery allowed	No signature allowed	APIs Payload (electronic manifest data)	Product Code	Product Name
03		X				"authority_to_leave": false, "allow_partial_delivery": false, "safe_drop_enabled": false,	00093	Parcel Post + Signature
							00096	Express Post + Signature
							00065	Parcel Post Return
							00068	Express Post Return
08			X			"authority_to_leave": true, "allow_partial_delivery": false, "safe_drop_enabled": false	00093	Parcel Post + Signature
							00096	Express Post + Signature
							00065	Parcel Post Return
							00068	Express Post Return
45		X		X		"authority_to_leave": false, "allow_partial_delivery": true, "safe_drop_enabled": false	00093	Parcel Post + Signature
							00096	Express Post + Signature
15			X	X		"authority_to_leave": true, "allow_partial_delivery": true, "safe_drop_enabled": false	00093	Parcel Post + Signature
							00096	Express Post + Signature
50	X					"authority_to_leave": false, "allow_partial_delivery": false, "safe_drop_enabled": true,	00093	Parcel Post + Signature
							00096	Express Post + Signature
51	X			X		"authority_to_leave": false, "allow_partial_delivery": true, "safe_drop_enabled": true	00093	Parcel Post + Signature
							00096	Express Post + Signature
09				X	X	"authority_to_leave": true, "allow_partial_delivery": true, "safe_drop_enabled": false	00091	Parcel Post (Non-Signature)
							00087	Express Post (Non-Signature)
49 *		X				"authority_to_leave": false, "allow_partial_delivery": false, "safe_drop_enabled": false IDENTITY_ON_DELIVERY feature must be used with an id_capture_type value of "addressee".	00093	Parcel Post Signature (Wine)
81		X				"authority_to_leave": false, "allow_partial_delivery": false, "safe_drop_enabled": false	00093	Parcel Post Signature (Wine)
82			X			"authority_to_leave": true, "allow_partial_delivery": true, "safe_drop_enabled": false,	00093	Parcel Post Signature (Wine)
83	X					"authority_to_leave": false, "allow_partial_delivery": false, "safe_drop_enabled": true	00093	Parcel Post Signature (Wine)

* Liquor Control Victoria require all wine and alcohol deliveries sent to a ‘first time’ customer or as a gift in Victoria to be delivered in-person to the addressee only. Australia Post has specific guidelines and packaging requirements for parcels containing wine and alcohol. Our [Wine Parcel Guide](#) and [Parcel and Freight Preparation Guide](#) outline these requirements, which were designed to ensure the safety of customer parcels, our team members and customers, and to meet regulatory requirements.

The eParcel Check Digit

The eParcel check digit is the last digit of the article number, and it is there to ensure the validity of the article number.

For the eParcel barcode the EAN/UCC-13 check digit algorithm (also used for the SSCC barcode) is used to produce a human readable check digit which ensures that when the eye-readable line is entered manually, a check is made on the accuracy of data entry.

All characters in the Parcel Post\Express Post barcodes are numeric except for the Merchant Location ID which could be alphanumeric. Care should be taken to ensure that your check digit algorithm can accommodate both types of MLIDs.

Prior to the calculation, any alpha character in the barcode needs to be replaced by the second digit of the equivalent value in the ASCII character set.

Reference table for converting alpha characters into digits

Character	ASCII code	Character	ASCII code
A	65	N	78
B	66	O	79
C	67	P	80
D	68	Q	81
E	69	R	82
F	70	S	83
G	71	T	84
H	72	U	85
I	73	V	86
J	74	W	87
K	75	X	88
L	76	Y	89
M	77	Z	90

Note: the article numbers can be either 21 or 23 characters long (depending on if the MLID is 3 or 5 characters), the A.P. Article ID or Article Number is displayed in human readable format, for calculation of the check digit the first 20 (3 characters MLID) or 22 (5 characters MLID) characters of the human readable are considered.

Example 1

The following is an example of an eParcel check digit calculation procedure for a GS1-128 barcode with a 3 characters MLID (DJS) 019931265099999891DJS123456701000600204. The first 20 characters of the human readable part of the barcode are highlighted.

- **Step 1:** Replace all alpha characters with the last digit of the ASCII representation as per Table 9.
84312345670100060020
- **Step 2:** Starting with the first number on the right, add all alternate numbers. Multiply the result by three
84312345670100060020 so $0 + 0 + 6 + 0 + 1 + 7 + 5 + 3 + 1 + 4 = 27 \times 3 = 81$
- **Step 3:** Starting with the second number on the right, add all the alternate numbers.
84312345670100060020 so $2 + 0 + 0 + 0 + 0 + 6 + 4 + 2 + 3 + 8 = 25$
- **Step 4:** Add the results of Step 2 and Step 3 Total = 106
- **Step 5:** Add the number needed to bring the total to the next multiple of ten. In this case it is 4, which is the check digit.
If a result is divisible by 10, then the check digit is 0.

Example 2

Now let's do one calculation for a 5 characters MLID (1JDQ1) to ensure that we are always getting it right
0199312650999998911JDQ1200000101000935008

- **Step 1:** Replace all alpha characters with the last digit of the ASCII representation as per Table 9.
1481120000010100093500
- **Step 2:** Starting with the first number on the right, add all alternate numbers. Multiply the result by three
1481120000010100093500 so $0 + 5 + 9 + 0 + 1 + 1 + 0 + 0 + 2 + 1 + 4 = 23 \times 3 = 69$
- **Step 3:** Starting with the second number on the right, add all the alternate numbers.
1481120000010100093500 so $0 + 3 + 0 + 0 + 0 + 0 + 0 + 0 + 1 + 8 + 1 = 13$
- **Step 4:** Add the results of Step 2 and Step 3 Total = 82
- **Step 5:** Add the number needed to bring the total to the next multiple of ten. In this case it is 8, which is the check digit.
If a result is divisible by 10, then the check digit is 0.

Check Digit calculation - Sample code (VBA)

```
Private Function CheckDigit(MyArticleNumber)
Do Until i = Len(MyArticleNumber)
MyDigit = If(Asc(Mid(MyArticleNumber, Len(MyArticleNumber) - i, 1)) > 47 And Asc(Mid(MyArticleNumber,
Len(MyArticleNumber) - i, 1)) < 58, Mid(MyArticleNumber, Len(MyArticleNumber) - i, 1),
Right(Trim(Str(Asc(Mid(MyArticleNumber, Len(MyArticleNumber) - i, 1)))), 1))
Mynumber = Mynumber + If(i Mod 2 = 0, 3 * (Val(MyDigit)), (Val(MyDigit)))
i = i + 1
Loop
CheckDigit = (Right(Str(10000 - Mynumber), 1))
End Function
```

Symbol Check Character

The symbol check character allows a quality check of the whole barcode, it is modulo 103 and it appears immediately before the stop character in the barcode.

It is not to be confused with the Parcel Post\Express Post Check digit which is used to validate the article number (data), They are two different things, unless you are writing software to assemble a barcode from scratch you do not need to know about the Symbol Check character as it will be automatically added by the barcode generation library or software in use.

Options for label printing

1) Shipping & Tracking APIs

It is recommended the use of the Australia Post Shipping & Tracking APIs for the following reasons:

- Developers are not required to acquire knowledge about barcodes and label formats as these are obtained by JSON messaging.
- It eliminates the need for changes to label formats to align to Post requirements when new products and services are added to the merchant's contract.
- The Shipping & Tracking APIs can manage the allocation of consignment and article numbers preventing duplication.
- The Shipping & Tracking APIs perform address validation (adds DPID) for successful delivery.
- The Shipping & Tracking APIs perform validation against the merchant contract and only allows printing of labels for products and services included in the merchant contract, it allows matching of electronic data

Note: further information about the Shipping and Tracking APIs can be obtained in the Australia Post Developers Centre (<https://developers.auspost.com.au/apis/shipping-and-tracking/reference>)

2) Zebra Printing Language (ZPL)

This technique is compatible with the label requirements and if used correctly is likely to achieve the desired outcome (quality barcodes and format). Before embarking on development, the merchant needs to be sure that the firmware of the hardware is compatible, particularly considering that older thermal printers can't generate 2D Datamatrix Barcodes.

Below are two examples of ZPL commands that result in correct barcode formats for both the GS1 128 as well as the GS1 2D Datamatrix Barcode.

```
^XA
^FO90,200^BY2^BCN,150,Y,N,Y,N
^FD>;>8019931265099999891>62NM>5100001901000930201^FS
^XZ
```

Generates the GS1 128 Linear barcode 0199312650999998912NM100001901000930201



```
^XA
^FO600,100^BXN,6,200,,,,
^FD_1019931265099999891JDQ77777902000931506_14202000_18008170314000000^FS
^XZ
```

Generates the GS1 2D Datamatrix Barcode 019931265099999891JDQ77777902000931506|4202000|8008170314000000



Label and barcode testing

Before using labels and barcodes produced using non-Australia Post systems, the following testing process must be followed:

- It is required that a series of labels be produced with various combinations of services to demonstrate the ability to calculate correctly the check digit and deal with different sizes of consignments.
- These labels must be forwarded to your eCommerce Integration Specialist, who will perform a quality test and ensure that they conform to the standards and can be read by Australia Post's scanners.
- A subsequent live test on the automatic sorter will confirm whether labels are suitable for automated sorting.

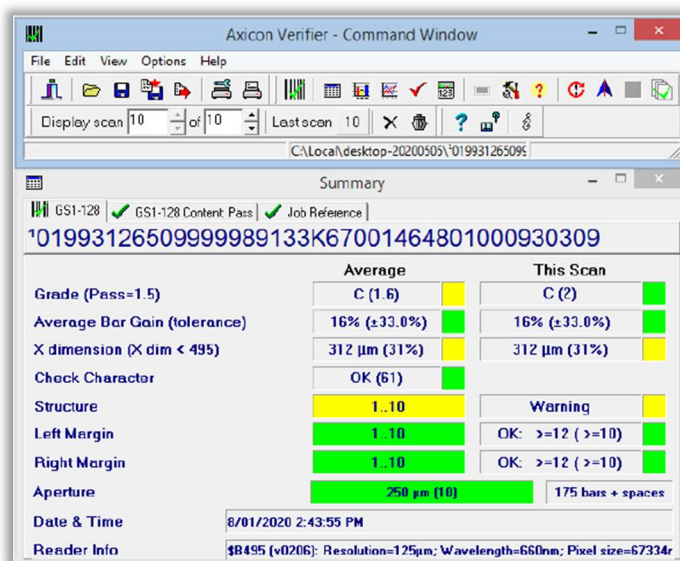
On completion of this test, your eCommerce Integration Specialist will provide you with written approval to produce labels in the production environment.

Testing for quality with the barcode Verifier

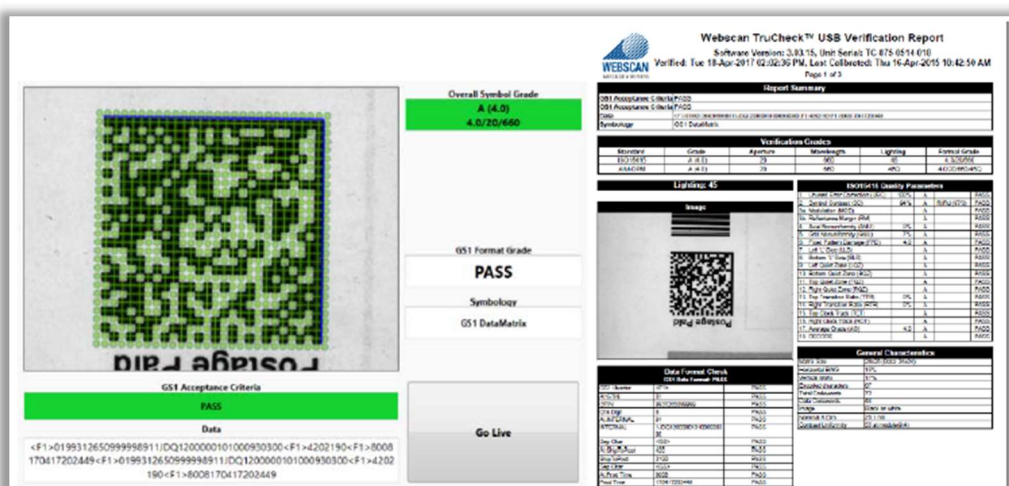
We use Axicon verifier equipment for quality checks on both the linear barcode and the 2D Datamatrix Barcode.

This equipment allows us to determine if barcodes are GS1 compliant and are suitable for automated processing.

In the example the barcode is well formed, and has passed the quality test, note the grade average (1.6) just above the minimum for the industry standard (1.5). This causes for the barcode to be rated as "C" (minimum rate for automated parcel processing).



The GS1 2D Datamatrix barcode tested below is both, GS1 and eParcel compliant



SSCC barcodes

Australia Post strongly recommends the use of the Parcel Post/Express Post standard barcode due to its built-in features which maximises the functionality available, minimises service compliance failures, supports consignment management, and facilitates automatic sorting of parcels through Australia Post Large Parcel Sorting Machines. This barcode also provides identification of the account or location from where items are sent without the need for operators to refer to back-end systems to find this information.

Australia Post will, in certain circumstances, accept other barcoding and numbering systems such as GS1 SSCC (Serial Shipping Container Code), provided the following is considered:

- A prefix (GS1 Company number followed by the first few characters of the SSCC number) is set aside for exclusive use on Parcel Post articles.
- A separate prefix (GS1 Company number followed by the first few characters of the SSCC number) is set aside for exclusive use on Express Post articles. This ensures Australia Post will be able to identify products being despatched by the merchant even if the electronic Order (manifest data) is not available.
- Set-up/implementation of a new account/sub-account to use SSCC could take up to three weeks as a new 'barcode mask' is required to be established with a new barcode number range for each new account. The new barcode mask then needs to be downloaded to all driver scanning devices and supporting systems so that these barcodes are identified as eParcel consignment when the barcodes are scanned.
- Generate an additional Datamatrix Barcode on the label for use for automatic sorting.
- Allocated MLID to be included as a prefix to the consignment number.
- A barcode mask is used to determine the type of product and service being provided since a product code and service code cannot be embedded within the barcode for merchants NOT using the eParcel barcode.
- Consultation with an eCommerce Integration Specialist is required to ensure that Australia Post systems can be configured to handle the merchant's barcode and numbering systems and to ensure that the data integrity of Australia Post systems is maintained.

Barcode strings required when using SSCC barcodes:

GS1 128 linear barcode: <F1>00123456789123456789 (Just the SSCC number including AI 00)

GS1 2D Datamatrix Barcode: <F1>019931265099999891**100123456789123456789**<F1>4203195<F1>8008200625101021

(SSCC in bold, in exactly in the same position where a standard Parcel Post \Express Post article Id would go).

Note: For the returns services, the use of the Australia Post Parcel Post/Express Post standard barcode is mandatory. SSCC barcodes **cannot be used** for eParcel Returns.

International labels

Australia Post requires that International barcoded labels to be sourced from the "Shipping and Tracking APIs" ("Create Label"/ "Get Label").

Customers are not permitted to generate International labels in house.

Preliminary “in house” barcode testing

To speed-up barcoded label development and code debugging, it is recommended for merchants and developers to use barcode testing tools and apps available online. Below is a list of some example apps that can be downloaded free of charge for use with smart phones

- GS1 Barcode Scanner (Android Only)
- Scandit (Apple and Android)

Following there is an example of the correct readings:

2D Datamatrix barcode - this reading was obtained from scanning the 2D barcode in figure 6 of this document with the NEOREADER app.

Check out what I just scanned with NeoReader

Check out what I just scanned with NeoReader:
"jd2019931265099999891JDQ0194571010009303085420219058008170417155407"

Note "jd2" at the start of the string, it is an indication that the FNC1 character is present.
Note also the diagonal line within a square just before application identifiers "420" and "8008", it indicates that the FNC1 is present as separator character.
By the way in this case DPID could not be determined hence why AI "92" and its corresponding FNC1 representation were not included in the barcode string.

To start scanning barcodes with your mobile phone, check out <http://www.neoreader.com> or download NeoReader from your app store.

NEOREADER allows you to email the results and this is a convenient feature as often the square symbols are only visible when opening the email attachment, so it is a good practise to email the results to yourself.

Note also "jd2" is a standard representation of the Function1 character when read by scanning devices (including the NEOREADER app). This does not mean the "jd2" is to be inserted in the barcode string, the same applies to those function 1 characters acting as AI separators. Depending on the library or technique in use, to achieve a function1 character, you need to check the documentation or contact the providers of your barcode tool for advice.

Check out what I just scanned with NeoReader:
"jd2019931265099999891JDQ019435001000930200542008415928843311258008170329183448"

GS1 128 Linear barcode - this reading was obtained from scanning the linear barcode in figure 5 of this document with the NEOREADER app.

JC1019931265099999891JDQ019457101000930308

Note the "JC1" in the linear barcode indicates that a FNC1 character is present in that position.

Note: "JC1" is a standard representation of the Function1 character on linear barcodes when read by scanning devices. For developers, **this does not mean** the "JC1" is to be inserted in the barcode string. Depending on the library or technique in use, to achieve a function1 character, you need to check the documentation or contact the providers of your barcode tool/software for advice.

Below are some examples of invalid 2D symbols when read by the NEOREADER app (note these issues preventing barcodes from being recognised as Australia Post barcodes affecting processing and tracking as a result).

Check out what I just scanned with NeoReader:

"\$019931265099999891QCZ000000101000600200\$4202141\$8008170213053042"

- Issue: Invalid Function1 character, there is a "\$" in the position of the Function1 characters

Check out what I just scanned with NeoReader:

"jd2019931265099999891336GN00010060100093020742020208008170331090609"

- Issue: characters are correct but located in the incorrect position, when acting as separator characters these should be before "420" and "8008" in the string below

Check out what I just scanned with NeoReader:

"_1019931265099999891KHH075306201000600207_14200851_18008170404163016"

- Issue: An underscore followed by a 1 "_1" is not a valid FNC1 character

Check out what I just scanned with NeoReader:

"jd20199312650999998917JHABCDEF0000159100093080204208000920000000008008170328180613"

- Issue: Two issues found, the article number is incorrect and the DPID is "00000000", if unknown this value and its application identifier are to be removed from the barcode string...also Postcode is only 3 characters long (should be 4)

Check out what I just scanned with NeoReader:

"jd2019931265099999891JDQ7777779020009315060420800817031 4000000"

- Issue: Space in the date and time

Check out what I just scanned with NeoReader:

"019931265099999891M4M3000017010009315060420421708008170308131228"

- Issue: Incorrect value as FNC1 at start of the string (should be "jd2" instead).

Check out what I just scanned with NeoReader:

"019931265099999891THK10015280200093030142050618008170308155758"

- Issue: No FNC1 characters at all

Check out what I just scanned with NeoReader:

"FNC1019931265099999891DJS123456701000600204FNC14203207FNC192FNC18008170303113922"

- Issue: String "FNC1" in the position of a FNC1 character is not valid

Check out what I just scanned with NeoReader:

"jd2019931265099999891JDQ0371679010006002090420232208008170302144700"

- Issue: Note two FNC1 characters before AI 8008 (If no DPID is available AI92 and its corresponding FNC1 character need to be removed).

Check out what I just scanned with NeoReader:

"019931265099999891JDQe42031499283279523e800820170216122430"

- Issue: No FNC1 at the beginning of the string, also incorrect separator character

Regular label quality checks and damaged labels

- a. Check the print quality is good – the ink should be black, not grey.



Not acceptable

- b. Check the barcode background colour – it should be all of the same colour, with no contrast.



Not acceptable

- c. Check for damage – ensure your barcode isn't torn or ripped in any way.



Not acceptable



- d. Check your printer heads are clean – this prevents the printing of vertical lines through the barcode.



Not acceptable

Label placement issues

This is critical to ensure address and barcodes are exposed to scanning devices during processing and delivery.

- a.** Check the barcode isn't obstructed – it should be clearly visible with nothing covering it eg plastic strapping or tape.



Not acceptable



Not acceptable

- b.** Check the barcode is visible on one side of the parcel only – ensure it's not wrapped around the edge of the parcel.



Not acceptable

- c.** Check the barcode is flat – it should be positioned on the parcel's largest part and avoid placing it over the packaging fold or flap.



Acceptable

Tracking details via API

Merchants generating labels in house are required to manage the allocation of tracking numbers ensuring these are not duplicated.

This section shows where to include tracking_details in the payload (electronic data) and in which format.

The example below shows a typical "Create Shipment" Json payload, however, it is also applicable to the "Create Order including Shipments" API.

Create Shipment Payload (with tracking details in light blue)

```
{
  "shipments":[
    { "from":{
      "name":"C Sender",
      "lines":[
        "1 Jedda Road",
        "suburb":"PRESTONS",
        "state":"NSW",
        "postcode":"2170"},
      "to":{
        "name":"Carol Receiver",
        "lines":[
          "8 Test Close"],
        "suburb":"SOUTH MORANG",
        "state":"VIC",
        "postcode":"3752"
      },
      "items":[
        {
          "product_id":"3D85",
          "length":"5",
          "height":"5",
          "width":"5",
          "weight":"1.5",
          "authority_to_leave":true,
          "allow_partial_delivery":false,
          "safe_drop_enabled":true,
          "tracking_details":{
            "article_id":"QPW500000201000930800"
            "consignment_id":"QPW5000002",
            "barcode_id":"019931265099999891QPW500000201000930800|4203752|8008200618120000"
          }
        }
      ]
    }
  ]
}
```

- Where the consignment_id is the MLID followed by seven digits
- The article_id is the consignment number followed by parcel count, product code, service code, postage paid indicator and check digit (see below in yellow)
- The barcode_id is the representation of the 2D barcode string on the label and includes all components below.

Component	Description
019931265099999891	is a constant prefix that indicates the barcode is an Australia Post one (fixed value)
QPW	Merchant location Id (MLID)
5000002	Consignment suffix (counter) ...from 0000001 to 9999999 per account before reusing numbers.
01	Parcel count within the consignment (maximum count is 20 per consignment)
00093	Product Code (see page 21)
08	Service Code (see page 21)
0	Postage paid (fixed value)
0	Check digit of the article_id (resulted from check digit algorithm on page 22)
	Pipe character required in the electronic data to represent the position of a Function1 character acting as group separator character in the 2D barcode (physical label).
420	GS1 Application Identifier for domestic postcode (fixed value)
3752	Destination postcode
	Pipe character required in the electronic data to represent the position of a Function1 character acting as group separator character in the 2D barcode (physical label).
8008	GS1 Application Identifier for date and time (fixed value)
200618120000	Date time of label generation ("YYMMDDhhmmss")

Users of SSCC numbers (Serial Shipping Container Codes) as Parcel Post / Express Post numbers need to include these in the article_id and barcode_id in the electronic data.

We'll modify the above example to illustrate how this is done...if the article number we use is SSCC barcode number (00)123456789123456789 then the tracking details in the electronic "Order" (manifest) data need to go as below:

```
"tracking_details":{
  "article_id":"00123456789123456789",
  "consignment_id":"QPW5000002",
  "barcode_id":"01993126509999989100123456789123456789|4203752|8008200618120000"
}
```

Please refer to the Australia Post Developers Centre for further details on the payload structure and requirements

<https://developers.auspost.com.au/apis/shipping-and-tracking/reference>

[End of Document]